RAG based pdf bot

The RAG based pdf bot is a query tool to search for information in the PDF files under the pdf folder.

The current query string is

"What is the Microphone sensitivity?"

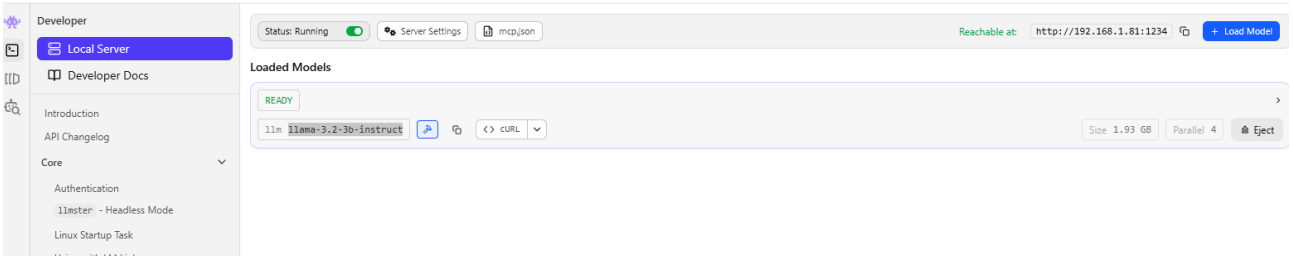
The PDF file, manual.pdf, is the document that contains the query result.

The steps to launch the pdf bot to make the query:

1. GIT clone the repo
   git clone https://github.com/haidong1965/pdf_bot_demo

2. Enter the repo and initialize python environment
   cd pdf_bot_demo
   python -m venv my_rag_project_venv
   my_rag_project_venv\Scripts\activate.bat
   pip install chromadb google-genai langchain_community pypdf sentence-transformers

3. Start the local LLM model from LM studio
4. Load LLM module llama-3.2-3b-instruct
5. Start local server



6. Start the pdf bot to make the query
   python complete_rag_pdf_bot.py
7. The query result is shown in the prompt

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits
Loading weights: 100%|                                                                                | 199/199 [00:00<00:00, 2613.75it
? Local Embedding Model 'BAAI/bge-small-en-v1.5' loaded successfully.
? ChromaDB Client initialized successfully.


########## STARTING INDEXING PHASE ##########

[STAGE 1/4] Found 1 PDF files to process.
   -> Successfully loaded 10 pages from manual.pdf

[STAGE 2/4] Chunking Documents...
? SUCCESS: Created 9 manageable chunks.

[STAGE 3/4] Generating Embeddings...

--- Generating 9 Embeddings using BGE Model ---
? Embedding generation complete.

[STAGE 4/4] Storing Data...
? SUCCESS: All data written to ChromaDB with unique IDs.

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? INGESTION PIPELINE COMPLETE! Data is indexed.
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########## STARTING QUERY PHASE ##########
[Phase 1/3] Database connection established to 'rag_knowledge_base'.
[Phase 2/3] Query Embedding generated successfully.

[Phase 3/3] Querying Knowledge Base...
? Phase 3 Success: Context retrieved successfully.

[--- Calling LM Studio API ---]

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? RAG PIPELINE EXECUTION FINISHED ?
FINAL QUERY: What is the Microphone sensitivity?
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Based on the provided context, the microphone sensitivity is 24dBV/Pa.
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